

Executive Summary | GLP-1 & Zipf

What did analyzing 1,000 scientific abstracts teach us beyond the textbook?

Dataset and method. The original 1,000 unique-PMID GLP-1 abstracts yield 161,515 baseline tokens and 10,691 terms (161.515 tokens per document). Four cumulative conditions isolate punctuation, stopwords and stemming; titles are excluded.

Finding 1 - The law is approximate and range-dependent. The log-log curve looks nearly linear, yet residuals are curved and 4,434 terms occur once. The final 10% is a flat $CF=1$ plateau. Seeing that plateau explains why a textbook straight line is an idealization rather than a model of every observed rank.

Finding 2 - There is no context-free exponent. Baseline slope=-1.2982, intercept=5.1468, $k=1.2982$, $R^2=0.9708$, $\log_{10} RMSE=0.0975$. The first 10% has $k=0.9271$, $R^2=0.9967$, and best approximates $k=1$ under our bands. High R^2 alone cannot establish a power law; the constant tail even has zero RMSE without meaningful R^2 .

Finding 3 - Preprocessing changes the experiment. Removing stopwords reduces tokens by 32.4% and changes k to 1.2414. Stemming preserves those tokens but reduces vocabulary to 7,815, raising k to 1.3487. Preprocessing is part of the result, not merely preparation before analysis.

Finding 4 - Topic importance differs from retrieval discrimination. GLP-1 occurs in 929/1,000 abstracts (IDF 0.074); semaglutide in 85 (IDF 2.465). A central topic word can weakly distinguish documents in a focused collection. CF measures repetition; DF measures document coverage and determines IDF.

IR consequence. A frequent head and sparse tail motivate different treatment of long postings and dictionary overhead, careful stopword policy, compression and TF-IDF. We measured distributions, not retrieval quality: word removal and weighting still require evaluation on relevant queries. The single-domain, relevance-selected sample also constrains generalization.

Conclusion. Zipf is useful as an empirical guide to workload and weighting even when k differs from one. These results replace an assumed universal line with a measured, preprocessing-sensitive description. The full report provides tables, sources and reproducibility. Optional cross-domain comparison omitted.